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The Structural Barrier Theorem: the Algebraic Books Cannot Select $\pi/7$

Elimination ideal = 0 · PSLQ null with positive controls · transcendence of $\pi/7$

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Abstract

This monograph records the structural barrier theorem of the OPEN9 v19 campaign: the algebraic book tower of the corrected system cannot select the surviving brick $\delta_C = \pi/7$. The three-level null test proceeds from the exact algebraic layer (the elimination ideal of the book system in $\mathbb{Q}[\delta]$ is zero — a Gröbner computation), through the integer-relation layer (PSLQ over $[1, d, \dots, d^8]$ at $d = \pi/7$ on 50/100/200 digits returns nothing, while the controls recover the known intra-sector relations), to the transcendence layer ($\pi/7$ is transcendental, so no non-zero polynomial over $\mathbb{Q}$ can vanish at it).

L1 + L3 give the barrier: selection can enter only through the transcendental inventory $\{\pi, \ln(16/9)\}$ of the books — finite-amplitude monodromy or Berry holonomy. The data agree: the tick does not select the brick (non-monotone, minimum at $k = 12$), and the Berry closure +0.062% is an observation with a multiple-testing caveat. Protocol tables, corollaries, honest caveats and reproduction commands are included.

Keywords: structural barrier, brick selection, $\delta_C = \pi/7$, elimination ideal, PSLQ, transcendence, Lindemann–Weierstrass, machine theorem.

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1. Introduction

Tenth in the cycle, this monograph records the structural barrier theorem: the machine-proved statement that the algebraic book tower of the corrected system cannot select the surviving brick $\delta_C = \pi/7$. The result closes OPEN9 question No. 1 (machine brick_selection.py) with a three-level null test — algebraic elimination, PSLQ integer-relation search, and a transcendence argument — and thereby redirects the selection question to the two legitimate mechanisms outside the algebraic sector: finite-amplitude monodromy and Berry holonomy.

The theorem is the honest negative result of the brick programme, and the series records it with the same standing as the positive theorems: the books determine how the clock moves once δ is given (monographs No. 1, No. 2, No. 7), but nothing in the algebraic sector fixes δ itself. Every layer of the null test is reproducible, and the control experiments demonstrate that the method would have found relations had they existed.

2. Notation and Setting

The brick parameter δ enters the corrected system through the additive convention $\kappa(\delta) = 2 - \delta^2/2$; the books are the clock family $\tau^*(\delta) = 27/(4 - \delta^2)$ (monograph No. 2) and the ring normalization. The surviving brick $\delta_C = \pi/7$ belongs to the septimal spinor sector ($bC = \pi^2/98$, phase $\pi/7$); the question is which mechanism — if any — singles out this value.

Three levels are probed. L1 (algebraic closure): does the book system impose any polynomial condition on δ ? The tool is a Gröbner elimination in $\mathbb{Q}[\delta]$. L2 (integer relations): is $\delta_C = \pi/7$ related to the known transcendental constants of the programme by a rational polynomial of degree ≤ 8 ? The tool is PSLQ on 50/100/200 digits. L3 (transcendence): a classical theorem-based argument about what algebraic equations $\pi/7$ can satisfy at all.

3. Statement of the Theorem

Theorem 10 (structural barrier). The algebraic book sector of the corrected system leaves the brick free:

(L1) the elimination ideal of the book system in $\mathbb{Q}[\delta]$ is zero:

$$\langle \kappa(\delta) + \frac{\delta^2}{2} - 2, \tau^*(4 - \delta^2) - 27 \rangle = \{0\} \subset \mathbb{Q}[\delta] \quad (10.1)$$

— no polynomial condition on δ follows from the books (machine Groebner/lex computation);

(L2) PSLQ over the basis $[1, d, d^2, \dots, d^8]$ at $d = \pi/7$ on 50, 100 and 200 digits returns no relation; the method controls recover the known intra-sector relations ($\pi/7$ versus $7\pi/30$: both multiples of π ; $\ln(16/9) = 4\ln 2 - 2\ln 3$), so the negative result is not a precision artifact;

(L3, theorem) $\pi/7$ is transcendental (Lindemann–Weierstrass: π is transcendental, and division by the algebraic 7 preserves transcendence); hence no non-zero polynomial $P \in \mathbb{Q}[x]$ can have $\pi/7$ as a root;

(conclusion) L1 + L3 imply the structural barrier: the algebraic book tower by itself cannot select $\delta_C = \pi/7$. The transcendental inventory of the books is $\{\pi$ (the Berry-form κ_C and the spinor quanta $7\pi/30$), $\ln(16/9)$ (the hexcycle clock $6\ln(16/9)$)}; selection can enter only through finite-amplitude monodromy (questions No. 2/No. 8) or through Berry holonomy (`s4_berry_one_brick`).

4. Sketch of the Machine Proof

4.1. L1: the elimination ideal is zero

The machine assembles the book polynomials in $\mathbb{Q}[\delta]$ — $\kappa(\delta) + \delta^2/2 - 2$ and $\tau^*(\delta)(4 - \delta^2) - 27$ — and computes a lexicographic Gröbner basis with δ the last variable. The elimination ideal (the intersection with $\mathbb{Q}[\delta]$) is the zero ideal: both generators involve δ only in combination with the free clock variables, and no element of the ideal depends on δ alone. The books therefore constrain the clock given δ , but say nothing about δ — exactly the content of the brick-ladder theorem (monograph No. 2), now expressed in elimination language.

4.2. L2: PSLQ and the controls

PSLQ searches for integer relations among $[1, d, \dots, d^8]$ at $d = \pi/7$ with 50, 100 and 200 working digits: none exist at any width. Two controls validate the instrument: (a) the pair

$(\pi/7, 7\pi/30)$ — both are rational multiples of π , and PSLQ recovers the relation over $[1, \pi, \pi/7, 7\pi/30]$; (b) the pair $(\ln(16/9), \ln 2, \ln 3)$ — PSLQ recovers $4\ln 2 - 2\ln 3 - \ln(16/9) = 0$. The negative at $\pi/7$ is therefore a genuine absence, not a precision failure.

4.3. L3: transcendence

The Lindemann–Weierstrass theorem implies π is transcendental; $\pi/7$ is then transcendental as well (division by the algebraic nonzero 7). Consequently, for every non-zero $P \in \mathbb{Q}[x]$, $P(\pi/7) \neq 0$. Combined with L1, no algebraic equation the books could impose — even in principle, at any degree — can be satisfied by $\pi/7$: the barrier is structural, not a matter of degree or precision.

Table 10.1. The three-level null test (q1)

Level	Instrument	Outcome
L1 algebraic	Gröbner elimination in $\mathbb{Q}[\delta]$	elimination ideal = $\{0\}$
L2 integer relations	PSLQ, $[1..d^8]$, 50/100/200 digits	no relations; controls positive
L3 transcendence	Lindemann–Weierstrass	$\pi/7$ transcendental; no algebraic roots
Conclusion	L1 + L3	structural barrier: books cannot select $\pi/7$

5. Numerical Results

Table 10.2. PSLQ protocol and controls

Target	Basis	Digits	Result
$d = \pi/7$	$[1, d, \dots, d^8]$	50 / 100 / 200	no relation
control: $\pi/7$ & $7\pi/30$	$[1, \pi, \pi/7, 7\pi/30]$	50	relation recovered (both multiples of π)
control: $\ln(16/9)$	$[\ln(16/9), \ln 2, \ln 3]$	50	$4\ln 2 - 2\ln 3 - \ln(16/9) = 0$ recovered

The data agreement layer completes the picture: the tick obstruction does not select the brick (non-monotone, minimum at $k = 12/14$ — monograph No. 2), while the Berry screening closes $\ln(\tau^*)$ to $+0.062\%$ at δ_C (monograph No. 7). The latter is an observation with a multiple-testing caveat: one brick out of a scanned ladder was found to close the book, and the closure is recorded as compatibility, not as a selection principle.

6. Corollaries

Corollary 10.1 (the selection question is transcendental). Any future selection mechanism for δ_C must act through the transcendental inventory $\{\pi, \ln(16/9)\}$ of the books — via finite-amplitude monodromy (the DSS programme, questions No. 2/No. 8) or via Berry holonomy. Algebraic refinements of the books cannot contribute.

Corollary 10.2 (the honest standing of the brick). The septimal sector ($bC = \pi^2/98$, phase $\pi/7$) remains an independent input of the programme: everything the series proves about the clock holds for any δ on the ladder law, and the brick enters only through the Berry book closure. This is the honest division of labour between algebra and spinor data.

Corollary 10.3 (a falsifiable border). The barrier is itself falsifiable: a single polynomial relation over $\mathbb{Q}$ involving δ_C and the book constants — certified by any future machine — would breach $L1/L2$ and invalidate the theorem. The three-level protocol is reusable at every future tower enhancement.

7. Honest Caveats

$L1$ is exact for the book system as written (the clock family and the ring normalization); a future tower enhancement could in principle add new books, and the elimination must be re-run — the theorem is proven for the current algebraic sector. $L2$ is a bounded-degree, bounded-precision search: relations of degree > 8 or with atypically large coefficients would be missed; the controls bound the plausible sensitivity but do not prove completeness. $L3$ uses the transcendence of π ; any argument that replaces $\pi/7$ by an algebraic surrogate would void it. The multiple-testing caveat on the Berry closure is recorded in the verdict and in monograph No. 7: the $+0.062\%$ closure is compatibility evidence, not selection. Finally, the

barrier does not assert that δ_C is wrong — only that its justification lies outside the algebraic books.

8. Reproducibility

Table 10.3. Reproduction protocol for Theorem 10

Step	Command	Output
1	<code>python3 brick_selection.py</code>	results/q1_brick_selection.json (L1/L2/L3 null test)
2	<code>python3 brick_scan_tick.py</code>	results/brick_scan_tick.json (tick non-selection data)
3	<code>python3 s4_berry_one_brick.py</code>	Berry closure +0.062% (observation layer)
4	<code>python3 -m pytest tests/ -q</code>	58 passed (q1 tests included)

The q1 machine runs in seconds (deterministic, offline, BLAS = 1). The data correspond to commit 5b256cd; the self-contained copy lives in `einstein_direct/open9/machines/`.

References

- [1] Repository wild8highlander/choptuik_ac_bc: `einstein_direct/brick_selection.py`; `einstein_direct/results/q1_brick_selection.json` (OPEN9 v19, question No. 1).
- [2] Ibid.: `einstein_direct/brick_scan_tick.py`; `einstein_direct/results/brick_scan_tick.json` (tick non-selection).
- [3] Ibid.: `einstein_direct/s4_berry_one_brick.py` (Berry closure observation).
- [4] Ibid.: `einstein_direct/README_EN.md` §§29–30.
- [5] Monographs No. 2, No. 7 of the theorems12 cycle.
- [6] F. Lindemann. Über die Zahl π . Math. Ann. 20, 213–225 (1882); K. Weierstrass, Zu Lindemann’s Abhandlung Über die Ludolph’sche Zahl. Math. Ann. 52, 213–226 (1885) — the transcendence backbone of layer L3.